Experiment No. 6

Frequency Synchronization and Recovery using FLL Band-Edge Loop

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Aim

To study frequency synchronization and carrier frequency recovery using a Frequency Locked Loop (FLL) Band-Edge algorithm in a QPSK communication system implemented using GNU Radio and PlutoSDR.

Objective

The objective of this experiment is to estimate and correct carrier frequency offsets using an FLL Band-Edge block and observe its effect on synchronization, symbol recovery, and constellation stabilization.

Theory

In practical digital communication systems, the transmitter and receiver oscillators are not perfectly synchronized. This results in carrier frequency offset (CFO), causing constellation rotation and symbol decoding errors.

A Frequency Locked Loop (FLL) is a feedback synchronization technique used to estimate and compensate for carrier frequency errors. The FLL Band-Edge algorithm utilizes the spectral characteristics of pulse-shaped signals to determine frequency offsets and perform coarse carrier recovery.

After frequency correction, timing recovery and carrier phase synchronization are performed to achieve accurate symbol detection.

The synchronization chain used in this experiment is:

$$\text{AGC} \rightarrow \text{Matched Filter} \rightarrow \text{FLL} \rightarrow \text{Clock Sync} \rightarrow \text{Costas Loop}$$

The FLL performs coarse frequency correction while the Costas Loop performs fine carrier phase recovery.

Equipment Required

- ADALM-PLUTO SDR
- GNU Radio Companion (GRC)
- Antenna
- USB Cable
- Computer/Laptop

Software Requirements

- GNU Radio
- PlutoSDR Drivers
- libiio Support

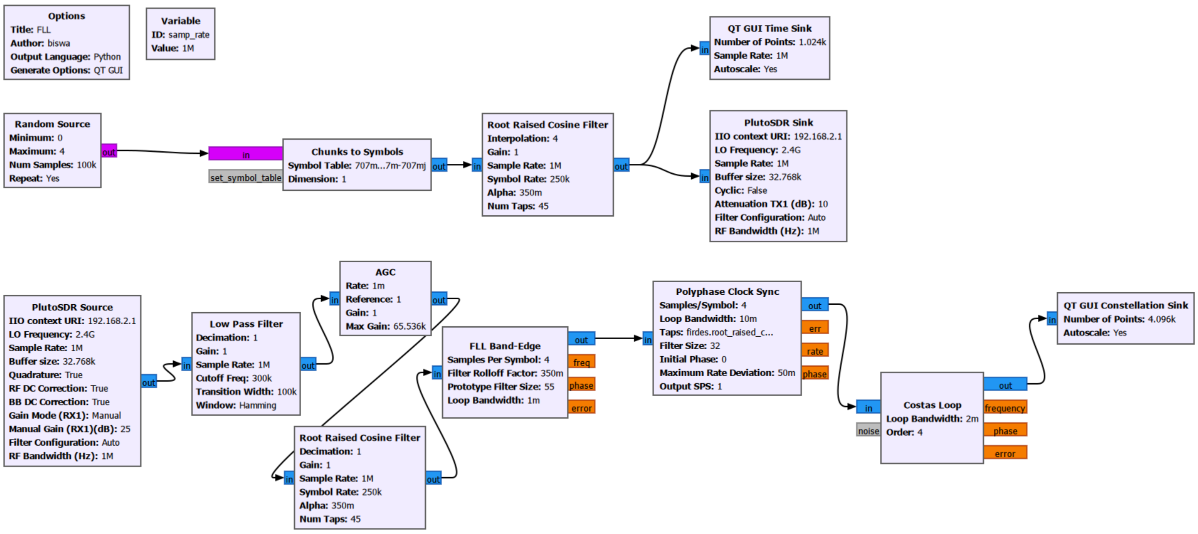

Figure 1: Frequency Synchronization using FLL Band-Edge, Clock Recovery and Costas Loop

Frequency Synchronization System

The above flowgraph implements a complete QPSK synchronization chain. The transmitter generates pulse-shaped symbols which are transmitted through PlutoSDR. At the receiver, AGC normalizes the signal, matched filtering improves signal quality, and the FLL Band-Edge block estimates and corrects carrier frequency offset. Timing recovery is performed using Polyphase Clock Sync, while the Costas Loop provides carrier phase synchronization before constellation recovery.

Flowgraph Description

Random Source

Generates random digital symbols for transmission.

Chunks to Symbols

Maps digital symbols into constellation points.

Root Raised Cosine Filter

Performs pulse shaping at the transmitter and matched filtering at the receiver.

PlutoSDR Sink

Transmits the pulse-shaped signal over the air.

PlutoSDR Source

Receives the transmitted signal.

Low Pass Filter

Removes unwanted frequency components and noise.

Automatic Gain Control (AGC)

Normalizes signal amplitude and compensates for channel variations.

FLL Band-Edge

Estimates and corrects carrier frequency offset between transmitter and receiver.

Polyphase Clock Sync

Performs symbol timing recovery and sampling synchronization.

Costas Loop

Removes residual carrier phase errors and stabilizes the constellation.

QT GUI Constellation Sink

Displays synchronized constellation points for analysis.

Working Principle

The transmitter generates random symbols and maps them into constellation points. These symbols are pulse-shaped using a Root Raised Cosine filter and transmitted using PlutoSDR.

At the receiver, the signal is filtered and normalized using AGC. The FLL Band-Edge block estimates carrier frequency offset and performs frequency correction. After frequency synchronization, the Polyphase Clock Sync block recovers symbol timing. Finally, the Costas Loop compensates for residual phase errors and produces a stable constellation suitable for symbol decoding.

Parameters

Parameter	**Value**
Sample Rate	1 MS/s
Symbol Rate	250 kSymbols/s
Samples per Symbol	4
Roll-off Factor (α)	0.35
Carrier Frequency	2.4 GHz
RRC Filter Taps	45
FLL Filter Size	55
Polyphase Filter Size	32
Costas Loop Order	4
RF Bandwidth	1 MHz

Table 1: System Parameters

Procedure

1. Open GNU Radio Companion.
2. Create the transmitter flowgraph using Random Source, Chunks to Symbols, and Root Raised Cosine Filter.
3. Configure PlutoSDR Sink for transmission.
4. Create the receiver flowgraph using PlutoSDR Source.
5. Add Low Pass Filter and AGC blocks.
6. Add a matched Root Raised Cosine filter.
7. Configure the FLL Band-Edge block for frequency synchronization.

8. Add Polyphase Clock Sync for timing recovery.
9. Add a Costas Loop for carrier phase synchronization.
10. Connect a QT GUI Constellation Sink.
11. Run the flowgraph and observe the synchronized constellation.

Observation

The FLL Band-Edge block successfully estimated and corrected the carrier frequency offset present in the received signal. After frequency correction, the Polyphase Clock Sync and Costas Loop achieved stable synchronization, resulting in a clear and properly aligned QPSK constellation.

Result

Frequency synchronization was successfully achieved using the FLL Band-Edge algorithm. The carrier frequency offset was corrected, enabling reliable timing recovery, carrier phase synchronization, and accurate constellation recovery.

Applications

- QPSK Communication Systems
- Software Defined Radio Receivers
- Satellite Communication
- Wireless Communication Systems
- Digital Modems
- Cognitive Radio Systems

Conclusion

The experiment demonstrated the use of a Frequency Locked Loop (FLL) for carrier frequency synchronization in a QPSK communication system. The FLL effectively corrected frequency offsets and improved receiver performance by enabling reliable timing recovery, carrier phase synchronization, and symbol detection.

References

1. GNU Radio Documentation
2. Analog Devices PlutoSDR Documentation
3. Digital Communication Systems by Simon Haykin
4. Software Defined Radio Principles and Applications